

DHEA, DHEAS and PCOS.

Approximately 20-30% of PCOS women demonstrate excess adrenal precursor androgen (APA) production, primarily using DHEAS as a marker of APA in general and more specifically DHEA, synthesis. The role of APA excess in determining or causing PCOS is unclear, although observations in patients with inherited APA excess (e.g., patients with 21-hydroxylase deficient congenital classic or non-classic adrenal hyperplasia) demonstrate that APA excess can result in a PCOS-like phenotype. Inherited defects of the enzymes responsible for steroid biosynthesis, or defects in cortisol metabolism, account for only a very small fraction of women suffering from hyperandrogenism or APA excess. Rather, women with PCOS and APA excess appear to have a generalized exaggeration in adrenal steroidogenesis in response to ACTH stimulation, although they do not have an overt hypothalamic-pituitary-adrenal axis dysfunction. In general, extra-adrenal factors, including obesity, insulin and glucose levels, and ovarian secretions, play a limited role in the increased APA production observed in PCOS. Substantial heritabilities of APAs, particularly DHEAS, have been found in the general population and in women with PCOS; however, the handful of SNPs discovered to date account only for a small portion of the inheritance of these traits. Paradoxically, and as in men, elevated levels of DHEAS appear to be protective against cardiovascular risk in women, although the role of DHEAS in modulating this risk in women with PCOS remains unknown. In summary, the exact cause of APA excess in PCOS remains unclear, although it may reflect a generalized and inherited exaggeration in androgen biosynthesis of an inherited nature. Copyright © 2014 Elsevier Ltd. All rights reserved.